

Luigi Casagrande

Materials and Nanotechnology Engineer

🇮🇹 Italian ✉ luigi.casagrande@outlook.it 🌐 luigicasagrande.it in luigi-casagrande

Education

- | | | |
|-----------|---|---|
| MS | POLITECNICO DI MILANO, Materials Engineering and Nanotechnology (<i>english</i>) <ul style="list-style-type: none"> • Master's Thesis in <i>Solid-State Aluminium Recycling</i> ★ Relevant coursework: "Metallic Materials and Product Innovation", "Composite Materials", "Life Cycle Assessment", "Materials for Energy" | Sep 2023 - Jul 2026
Milano, Italy 🇮🇹 |
| EX | SHANGHAI JIAO TONG UNIVERSITY, Materials Engineering <ul style="list-style-type: none"> • <i>World Wide Exchange Program</i> offered by Politecnico di Milano ★ Projects: "Physical aging in glassy polymers: stress relaxation and creep compliance analysis", "Molecular dynamics modelling of melting behaviour in pure Ni, Zr, and Ni-Zr alloys" | Sep 2024 - Feb 2025
Shanghai, China 🇨🇳 |
| BS | POLITECNICO DI MILANO, Materials Engineering and Nanotechnology (<i>italian</i>) <ul style="list-style-type: none"> • Bachelor's Thesis in <i>Printed Devices for Energy Storage</i> ★ Relevant coursework: "Steelmaking", "CAD Modelling", "Data Analysis and Experimental Methods", "Structural Mechanics" | Sep 2020 - Jul 2023
Milano, Italy 🇮🇹 |
| EX | KAWANA WATERS HIGH SCHOOL, STEM subjects <ul style="list-style-type: none"> • Independent study term abroad during High School | Jul 2016 - Sep 2016
Queensland, Australia 🇦🇺 |

Skills

Languages: • Italian (*native*) • English (*fluent*) • Chinese (*elementary*) • French (*elementary*)

Programming & Scripting: Python · $\LaTeX$ · JavaScript · C++ · HTML/CSS

Data Analysis & Productivity Tools: Microsoft Office (Word, Excel, PowerPoint, Access) · Power BI · JMP Pro

Engineering Design & Simulation Software: Autodesk Inventor · SolidWorks · Abaqus FEA · COMSOL Multiphysics

Technical & Scientific Computing Environments: Linux (Debian) · openLCA · LAMMPS · MATLAB

Certificates

- | | |
|--|-------------|
| IELTS Academic ➤ issued by: <i>British Council</i> | 2023 |
| English language certification, C1 Level (7.5) | |
| International Pool Lifeguard Certification ➤ issued by: <i>Federazione Italiana Nuoto</i> | 2017 |
| Pre-Academic Piano Certificates ➤ issued by: <i>Conservatori di Musica italiani</i> | 2013 & 2016 |
| Marine Military Sailing Course ➤ issued by: <i>Federazione Italiana Vela</i> | 2015 |

About Me

- Since childhood, I have pursued **competitive swimming** at a national level, alongside an active engagement in **music** through solo and ensemble performances. These sustained disciplines have shaped a highly determined, resilient, and **results-driven approach**.
- I have always **challenged myself** by participating in mathematics, computer science, and chemistry competitions, and was awarded the 'Aurea' Cup in the high school team championship. These experiences enhanced my **teamwork abilities**, **critical thinking**, and **problem-solving** skills.
- Travelling and engaging with different cultures have enhanced my **adaptability** and **open-minded approach** to diverse environments, contributing to my full proficiency in **English** and my ongoing study of the **Chinese** language.